

STM32WB Wireless Series

Bluetooth™ 5 & IEEE 802.15.4



Deliver best-in class IoT solutions with built-in key storage, OTA firmware updates and protocol concurrency control

A WIRELESS DUAL-CORE BRAIN

The STM32WB series is a dual-core, multi-protocol and ultra-low-power 2.4 GHz MCU system-on-chip.

It supports Bluetooth™ 5 as well as IEEE 802.15.4 communication protocols (in Single and Concurrent modes) covering a wide spectrum of IoT application needs.

Based on ST's best-in-class, ultra-low-power STM32L4 MCU, the STM32WB series reduces development time and BOM cost, extends application battery life and inspires innovation thanks to its rich and flexible peripheral set.

The STM32WB series is designed to fit industrial, healthcare and consumer applications.

BLUETOOTH™ 5 & IEEE 802.15.4

The STM32WB SoC offers multi-protocol stacks including Bluetooth™ 5, OpenThread and ZigBee® 3.0 as well as standardized IEEE 802.15.4 protocols in Concurrent mode for best-in-class RF performance and dedicated core to radio activity provides SW flexibility and better user experience.

IP PROTECTION

STM32WB devices offer device integrity and industrial IP protection features to meet manufacturers' increasing demand for brand protection.

Features	Benefits
Dual-core solution in a single die	Dual-core solution with independent clock trees ensures real-time RF execution and optimized PCB and BOM
TX: 5.2 mA, RX: 4.5 mA BLE: -96 dBm, 802.15.4: -100 dBm	Extended battery life time. Perfect fit for coin cell battery Comfortable and robust operating distance of connection
Integrated balun	Reduces BOM cost and PCB footprint
OTA firmware updates	Easy fleet maintenance
Crystal-less USB 2.0 FS interface	Optimized BOM cost. Battery charging detection
LCD driver, integrated booster	Only a simple low-cost glass display is needed
Quad-SPI XIP	Simple way to upgrade active memory on existing designs.
Customer key storage Secure bootloader	Offers brand protection, IP protection and device integrity

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